



POLITECNICO DI TORINO

Corso di Laurea in Cybersecurity

Tesi di Laurea

Iterative Approach for Network Security Automation

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Capitolo 1

Introduction

1.1 Context

Modern enterprise networks are characterized by growing complexity: hundreds of interconnected nodes, heterogeneous services, and security policies that must be consistently enforced across distributed infrastructure. Packet-filtering firewalls represent the primary mechanism for enforcing such policies, selectively allowing or blocking traffic flows based on predefined rules. However, as networks scale in size and dynamism, manually configuring firewalls becomes error-prone, time-consuming, and increasingly difficult to reason about in terms of formal correctness. An automated approach is therefore necessary — one that can guarantee both the correctness of the generated configuration and its optimality with respect to resource usage. It is in this context that VEREFOO was developed: a tool for the automated, formally correct allocation of firewalls within a network topology and the automatic generation of their filtering rules.

1.2 VEREFOO

1.2.1 Overview

VEREFOO (VERification REasoning on Firewall Optimization) is a tool designed to automate the allocation and configuration of packet-filtering firewalls in a network. It takes as input a network topology, a set of Network Security Requirements (NSRs) — high-level specifications of which traffic flows must be allowed or denied — and a Firewall Allocation Schema (FAS), which defines the candidate positions in the topology where firewalls may be placed. From these inputs, VEREFOO produces two outputs: an updated topology indicating which nodes have been allocated as active firewalls, and a vendor-independent configuration for each firewall specifying the filtering rules to be enforced. The tool guarantees three fundamental properties: *automation*, as the entire allocation and configuration process requires no manual intervention; *formal correctness*, as the generated configuration is proven to satisfy all specified NSRs; and *optimality*, as the number of allocated firewalls and generated rules is minimised. These guarantees are achieved by encoding the problem as a MaxSMT instance and solving it with the Z3 solver. The resulting configuration can further be translated into vendor-specific formats, currently supporting iptables, BPF, and OVS.

1.2.2 React-VEREFOO

Capitolo 2

Iterative React-VEREFOO

2.1 Idea of the Solution

2.2 Implementation

2.3 Example Run

Capitolo 3

Experimental Evaluation

3.1 Test Presentation

3.2 Test Configurations

3.3 Results

3.4 Comparison and Discussion

Capitolo 4

Conclusions

Capitolo 5

Executive Summary